A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Noise pollution - Set 1

Mark scheme

AQA spec 3.4.3.2.13; Pollution; textbook pp. 286-295

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** The human ear is sensitive to a very wide range of sound pressures / powers (about 10^{12} range) **1** A logarithmic scale compresses this into a convenient range of numbers: each 10 dB increase represents a ten-fold increase in sound intensity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Difference = 30 dB = 3 x 10 dB **1** = 10^3 = 1000 times **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** 88 dB **1** Doubling the sound intensity increases the level by 3 dB (logarithmic scale) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.4	1 mark for the point below. **1** 0 dB is set as the quietest sound that can be heard (at 1000 Hz) - the threshold of human hearing – quieter sounds exist but cannot be heard **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **threshold** → the concentration below which no harm is detected	1
01.5	**1 mark each** 1 mark for each of the 3 points below. **1** 20 to 20 000 Hz **1** dB(A) weights sounds between about 1000 and 4000 Hz more heavily **1** Because the nerve cells / hair cells in the cochlea are more sensitive to these frequencies so they are heard more easily / cause more annoyance – range narrows with age or noise damage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-287; AQA spec 3.4.3.2.13.	
02.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Stress-related illness:** ulcers, high blood pressure, heart disease **1 Behavioural changes:** irritability, aggression, lack of concentration leading to accidents **1** Communication problems / educational underperformance (eg schools under flight paths) **1** Sleep disturbance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 3 points below. **1 Acoustic fatigue:** objects exposed to sound at their natural resonant frequency vibrate, causing stress cracking / structural failure (eg early jet engine cowlings) **1 Vibration damage:** repetitive vibration from heavy vehicles damages buildings, bridges, underground pipes **1 Shock impacts:** the force of a sudden very loud noise (sonic boom) breaks windows / dislodges masonry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Birds that sing to establish territories / attract mates cannot be heard against background noise **1** Birds scared by sudden noise leave nests so predators take eggs / chicks – cliff-nesting seabirds knock eggs or chicks into the sea **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Whales / dolphins use hearing / echolocation to find food and communicate – loud sonar causes hearing damage if close to the source **1** Behavioural changes / disorientation possibly linked to mass strandings – reduced feeding success **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 287; AQA spec 3.4.3.2.13.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** A turbofan at the front forces bypass air around the inner cowling / core jet **1** The bypass air smooths the flow of the turbulent hot exhaust gases, reducing the turbulence that creates noise **1** The higher the ratio of bypass air to jet exhaust the quieter the engine (Trent 1000 quietest, Olympus noisiest) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
03.2	**Any 2 of 6** 1 mark each for any 2 of the 6 points below, in any order, maximum 2. **1** Chevron (serrated) nozzles mix bypass air with surrounding air more smoothly **1** Acoustic liners in the engine cowlings absorb noise **1** Hush kits fitted to older low-bypass engines **1** Fairings on undercarriage and wing-flap hinges reduce turbulence – welded rather than riveted panels **1** Lighter composite aircraft need less thrust **1** Blended wing with engines on top so the body shields the ground **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
03.3	**1 mark each** 1 mark for each of the 4 points below. **1** Constant (3°) descent avoids periods of high thrust in a stepped descent and keeps aircraft higher for longer **1** Night flights (especially take-offs, which are noisier) are restricted / banned when people are sleeping and background noise is low **1** Noisier aircraft pay higher charges / use up more quota points and fines for exceeding limits, so airlines are encouraged to buy quieter aircraft **1** More time to slow down after landing so wheel brakes can be used instead of noisy reverse thrusters **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quota** → a legal limit on the mass or number that may be caught **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	4
03.4	1 mark for the point below. **1** Engines run at higher power / maximum thrust and the steep climb angle needs full power (although the aircraft climbs quickly out of earshot) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 288-292; AQA spec 3.4.3.2.13.	
04.1	**1 mark each** 1 mark for each of the 4 points below. **1 Wheel vibration:** track polishing / grinding, sound-absorbing ballast, sound-absorbing suspension **1 Pantograph:** aerodynamic fairing **1 Wheel squeal:** lubrication of wheels / track on corners **1 Braking:** composite-material brakes instead of cast iron **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
04.2	1 mark for the point below. **1** Roads pass through / near almost all residential areas and traffic is continuous through the day, unlike airports or railways which affect limited areas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
04.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Sound-absorbing road surfaces (porous asphalt / rubber-crumb asphalt) **1** Acoustic fences, embankments / baffle mounds, walls or dense tree planting **1** Traffic management for free flow (less acceleration noise) – re-routing traffic away from residential areas **1** Double glazing / acoustic insulation of buildings **1 Vehicle design:** better aerodynamics, engine insulation, quieter exhausts, quieter tyres **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
04.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Air bubble curtains around the pile driver absorb noise **1** Acoustic survey to detect cetaceans first and delay work if present – increase volume gradually (soft start) so animals can leave **1** Drilling instead of pile-driving **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 292-294; AQA spec 3.4.3.2.13.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 5 points below. **1** Use a calibrated sound meter (dB(A)) at fixed distances (eg 10, 50, 100, 250, 500 m) from the track behind each barrier and at a control site with no barrier **1 Standardise the noise source:** same cars / same race or a standard test noise – same time of day, wind speed / direction, weather – same height of meter above ground **1** Take repeated readings at each distance and calculate means (and standard deviation) to assess variability **1** Use a measure such as LA10 or Leq over a set period rather than a single peak reading **1** Compare reduction in dB relative to the control at each distance – plot noise level against distance for each barrier – statistical test (t-test) to assess significance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **standard deviation** → a measure of the spread of the data about the mean • **t-test** → a test for a difference between the means of two normally distributed sets **Do not credit** • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	5
05.2	1 mark for the point below. **1** Wind carries sound / affects how far it travels so different wind directions would change readings independently of the barrier **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** LA10 is the noise level exceeded for 10% of the period - a measure of the noisiest periods **1** LA90 is the level exceeded for 90% of the period - a measure of the quietest / background level **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2

Do not write outside the box

Question	Marking guidance	Marks
05.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Double / triple glazing **1** Acoustic insulation in walls / roof **1** Bedrooms on the side facing away from the track **1** Dense hedges / trees and earth mounds in gardens **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 293-295, 418; AQA spec 3.4.3.2.13.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Principles:** dB is logarithmic (10 dB = 10x; 3 dB = 2x) so small reductions in dB are large reductions in intensity – effects on humans (hearing, stress, behaviour) and wildlife (breeding birds, bats, cetaceans) • **Aircraft design:** high bypass ratio engines, chevron nozzles, hush kits, acoustic liners, fairings, composites, blended wing • **Aircraft operation:** take-off angle, flight paths avoiding urban areas, constant descent, reduced reverse thrust, night restrictions, Chapter noise limits, fines and noise charges, quota count • **Airport design / location:** away from population centres, taxi and engine-test areas away from housing, acoustic walls, baffle mounds, land-use restrictions, free acoustic insulation • **Railway:** track polishing, sound-absorbing ballast and suspension, pantograph fairings, wheel lubrication, composite brakes • **Road:** porous / rubber asphalt, acoustic fences, embankments, tree planting, aerodynamics, engine insulation, traffic management and routing, double glazing • **Industrial:** silencers on compressors, pressing instead of stamping, nylon rollers, drilling instead of pile-driving, bubble curtains, timing of blasting, acoustic mats / curtains, remote operation, ear protection • **Domestic:** acoustic absorbers in appliances, volume limiters, selection of quieter appliances, ear defenders • **Evaluation:** which approaches are most effective / cost-effective – conflicts (airports near cities for convenience) – legislation and monitoring (Leq 57 dB contours) needed to enforce **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9
	• Any appropriate named example, including ones that are not listed here. • **quota** → a legal limit on the mass or number that may be caught **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	
	Model answer - Level 3 (9 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Because the decibel scale is logarithmic, a 10 dB cut is a tenfold reduction in intensity and 3 dB halves it, so modest-sounding design changes matter. Aircraft engines have been made quieter by high-bypass-ratio turbofans, chevron nozzles, hush kits and acoustic liners, with fairings and composite airframes reducing airframe noise. Operation can be changed too: steeper take-off angles and flight paths that avoid housing, constant-descent approaches, restricted reverse thrust, night curfews, Chapter noise limits enforced by fines and quota counts. Airport location and layout - away from population centres, engine-test areas and taxiways kept from housing, acoustic walls and baffle mounds, land-use planning and free insulation for residents - deal with the noise that remains.	
	Railways use track polishing, sound-absorbing ballast and suspension, pantograph fairings, wheel lubrication on curves and composite brakes. Roads use porous or rubber-crumb asphalt, acoustic fences and embankments, tree planting, engine insulation and aerodynamics, traffic routing and double glazing. Industry fits silencers to compressors, presses rather than stamps, uses nylon rollers, drills instead of pile-driving, surrounds piling with bubble curtains to protect cetaceans, times blasting, hangs acoustic mats and operates remotely, with ear protection as a last line. Domestic appliances use acoustic absorbers and volume limiters.	
	Design changes at source are the most effective because they reduce noise for everyone; operational limits depend on enforcement; location works only for new developments, and airports are still sited near cities for convenience. Legislation and monitoring of noise contours (57 dB L_{eq}) are what turn these options into actual reductions.	
	Weaker answer - Level 2 (5 marks)	
	Noise pollution can be reduced in different ways. Aircraft can be designed with quieter engines and can fly at night less often. Airports can be built away from cities. Roads can have sound barriers next to them and quieter road surfaces. Trains can have their tracks polished and use better brakes. Factories can use quieter machines and workers can wear ear defenders. Noise can also be reduced by planting trees. These methods reduce the noise that reaches people.	
	What makes the difference	
	• Level 2 gives correct but generic examples ('quieter engines'); Level 3 names the design features (high bypass ratio, chevron nozzles, acoustic liners) the question's 'named examples' requires.	
	• Level 3 organises by design / operation / location as the question asks; Level 2 mixes them.	
	• The decibel-scale point and the closing evaluation of which approaches work best lift Level 3 into the top band.	
06.2	1 mark for the point below. **1** The area within which the average (equivalent continuous) sound level between 07:00 and 23:00 in summer exceeds 57 dB - taken as the onset of significant community annoyance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-295; AQA spec 3.4.3.2.13.	

Do not write outside the box